

MODULE-4:

This document is the student guide for Module 4 of the AWS Academy Cloud Architecting course, titled "Adding a Storage Layer with Amazon S3". This module provides a comprehensive overview of Amazon S3, its features, benefits, use cases, and best practices for designing secure and efficient storage solutions. The following is a detailed summary of the important points covered in the document:

Understanding Amazon S3

- **Types of Storage:** The document begins by differentiating the three basic types of storage:
 - **Block Storage:** Data is stored in fixed-sized blocks on a device. This type of storage is commonly used for operating systems and databases.
 - **File Storage:** Data is organized in a hierarchical file structure, similar to a network drive.
 - **Object Storage:** Data is stored as objects, each with a unique identifier (key), data, metadata, and a version ID. **Amazon S3 is an object storage service.**
- **Amazon S3 Features:**
 - **Virtually Unlimited Storage:** S3 can store vast amounts of unstructured data.
 - **Buckets and Objects:** Data is organized into buckets, which act as containers for objects.
 - **Object Structure:** Objects consist of data, metadata (descriptive information about the data), an object key, and a version ID.
 - **Global Unique URLs:** Each object has a globally unique URL.
- **Benefits of Amazon S3:**
 - **Durability:** Data is highly durable, with 11 nines of durability (a 0.000000001% chance of data loss per year).
 - **Availability:** S3 provides high availability, ensuring data accessibility.
 - **High Performance:** S3 delivers high performance for data uploads and retrievals.
 - **Scalability:** S3 can automatically scale to handle varying storage demands.

Use Cases for Amazon S3

- The document outlines several common ways customers use Amazon S3, including:
 - **Media Hosting:** Storing and distributing videos, photos, music files, and other media content.
 - **Hosting Static Websites:** Hosting websites composed of HTML, CSS, JavaScript, images, and other static content.
 - **Data Store for Computation and Analytics:** Storing data used for analysis, processing, and big data workloads.
 - **Backup and Archive Critical Data:** Backing up and archiving data for disaster recovery and long-term storage.

Transferring Data to and from Amazon S3

- **Uploading Objects:** There are several options for uploading objects to Amazon S3:
 - **AWS Management Console:** A web-based interface for simple uploads (up to 160 GB).
 - **AWS CLI:** A command-line tool for uploading and downloading objects.
 - **AWS SDKs:** Programming libraries for integrating S3 uploads and downloads into applications.
 - **Amazon S3 REST API:** A programmatic interface for interacting with S3.
- **Multipart Upload:** This feature allows you to upload large objects in parts, improving throughput and resilience.
- **S3 Transfer Acceleration:** This feature leverages Amazon CloudFront's global network to speed up data transfers to and from S3 buckets.
- **AWS Transfer Family:** A managed service that enables secure file transfers to and from S3 and Amazon EFS using protocols like SFTP, FTPS, and FTP.

Storing Content with Amazon S3

- **Object Storage Classes:** Amazon S3 offers different storage classes to meet varying data access and cost requirements:
 - **S3 Standard:** For frequently accessed data.
 - **S3 Intelligent-Tiering:** Automatically moves data to the most cost-effective tier based on access patterns.
 - **S3 Standard-IA (Infrequent Access):** For infrequently accessed data.
 - **S3 One Zone-IA:** A lower-cost option for data stored in a single Availability Zone.
 - **S3 Glacier Instant Retrieval:** For long-lived, rarely accessed data that requires millisecond retrieval times.
 - **S3 Glacier Flexible Retrieval:** For data that does not require immediate access and can be retrieved within minutes to hours.
 - **S3 Glacier Deep Archive:** The lowest-cost storage class for long-term archival.
 - **S3 on Outposts:** Extends S3 to on-premises AWS Outposts environments.
- **Lifecycle Configurations:** You can define lifecycle rules to automatically transition objects between storage classes or expire objects after a specified period, optimizing storage costs.
- **Versioning:** S3 versioning allows you to preserve, retrieve, and manage multiple versions of an object.
- **Cross-Origin Resource Sharing (CORS):** Enables web applications hosted on different domains to access objects in your S3 bucket.
- **Data Consistency Model:** Amazon S3 provides **strong read-after-write consistency** for all operations, ensuring data integrity and simplifying application development.

Designing with Amazon S3

- **Default Security Configurations:**
 - **Private Buckets and Objects:** S3 buckets and objects are private by default, requiring explicit access permissions.

- **Default Encryption:** Objects are encrypted by default using server-side encryption with Amazon S3 managed keys (SSE-S3).
- **Encryption Options:** You can choose from various server-side and client-side encryption options to protect your data.
- **Access Control Tools:** Amazon S3 provides tools for managing access to buckets and objects:
 - **Block Public Access:** Prevents public access to buckets.
 - **IAM Policies:** Define permissions for users and roles.
 - **Bucket Policies:** Control access based on specific rules.
 - **Access Control Lists (ACLs):** Define granular permissions (less commonly used).
 - **S3 Access Points:** Provide named network endpoints with specific permissions for different applications.
 - **Preassigned URLs:** Grant temporary access to objects.
 - **AWS Trusted Advisor:** Checks for bucket permissions that grant global access.
- **Region Selection:** Consider factors such as data privacy laws, user proximity, service availability, and cost when choosing an AWS Region for your S3 buckets.
- **Amazon S3 Inventory:** This feature provides a list of objects and their metadata, helpful for auditing and managing your storage.
- **Cost Considerations:** Amazon S3 pricing is based on storage usage, data transfer, requests, and other factors. Understand the pricing model and leverage features like storage class transitions and lifecycle policies to optimize costs.

Applying the AWS Well-Architected Framework

- The document emphasizes the importance of applying the AWS Well-Architected Framework principles to your storage solutions, highlighting best practices for:
 - **Security:** Protect data at rest and in transit, enforce access controls, and utilize encryption.
 - **Reliability:** Design for fault tolerance and leverage S3's durability and availability features.
 - **Performance Efficiency:** Choose appropriate storage classes, utilize features like S3 Transfer Acceleration and multipart upload, and consider data access patterns.
 - **Cost Optimization:** Select cost-effective storage classes, implement lifecycle policies, and use S3 Inventory to analyze storage usage.

Hands-on Lab

- The module includes a lab where you create a static website hosted on Amazon S3, configure security settings, set up versioning, and define lifecycle policies.

Conclusion

This detailed summary provides an overview of the key points covered in the AWS Academy Cloud Architecting Module 4 student guide. For a deeper understanding, refer to the original source for specific examples, detailed explanations, and hands-on exercises.

